Hyunho Lee

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Research Interests

Remote sensing images based deep learning model for the water resources management

Spatiotemporal data mining for the water resources management

Spatial data science and geographic information science

Education

Ph.D. Geographic Information Science — **Aug. 2022 – Present**
Arizona State University — *Tempe, United States*
Advisor: Prof. Wenwen Li

M.S. Computer Science — **Mar. 2008 – Feb. 2010**
KAIST(Korea Advanced Institute of Science and Technology) — *Daejeon, Republic of Korea*
Thesis: The layer-based vector texture for 3D rendering
Advisor: Prof. Kwangyun, Wohn

B.E. Information and Computer Engineering — **Mar. 2001 – Aug. 2007**
Ajou University — *Suwon, Republic of Korea*

Publications

[1] J Park, **H Lee** (2020) "Prediction of high turbidity in rivers using LSTM algorithm". *Journal of Korean Society of Water and Wastewater* 34 (1), 35-43, https://doi.org/10.11001/jksww.2020.34.1.035

[2] J Kim, M Park, Y Yoon, **H Lee** (2020) "Application of recurrent neural network for inflow prediction into multi-purpose dam basin". *Advances in Hydroinformatics*, 397-408, https://doi.org/10.1007/978-981-15-5436-0_31

[3] J Park, **H Lee**, CY Park, S Hasan, TY Heo, WH Lee (2019) "Algal morphological identification in watersheds for drinking water supply using neural architecture search for convolutional neural network". *Water* 11 (7), 1338, https://doi.org/10.3390/w11071338

Work Experience

Senior Manager, Digital Water Platform Dept., *K-water* — Jan. 2021 – Jun. 2022

Manager, Digital Innovation Dept., *K-water* — Jan. 2020 – Dec. 2020

Manager, Data Center Dept., *K-water* — Jan. 2019 – Dec. 2019

Manager, Water Data Collection and Analysis Dept., *K-water* — Jan. 2018 – Dec. 2018

Manager, Human Resources Management Dept., *K-water* — Jan. 2013 – Dec. 2017

Staff, Information System Management Dept., *K-water* — Jul. 2010 – Dec. 2012

Teaching Experience

In-house Instructor, Data Analysis with R programming, *K-water* Apr.2018 – May.2021

Teaching Assistant, Introduction to Programming (CS101), *KAIST* Mar. 2009 – Jul.2009

Honors and Awards

1st Place Prize, *5th Bigdata analysis competition in K-water* Oct. 2021

Academic Conference Paper Award, *Korean Society of Environmental Engineering Annual Conference* Nov. 2020

12th place, *ACM-ICPC (International Collegiate Programming Contest) Asia-Seoul Regional* Nov. 2003

Certification

Advanced Data Analytics Professional, *certificated by K-Data, Republic of Korea* Apr. 2019
(Pass rate: 2.76%)

MOOC's – Online Certification

Machine Learning, CSMM.102x, *ColumbiaX (EDX)* Jan. 2017 – Apr. 2017

Mathematical Fundamentals for Data Science, *Korea University (K-MOOC)* Oct. 2017 – Jan. 2018

Artificial Intelligence and Machine Learning, *KAIST (K-MOOC)* Jan. 2018 – Feb. 2018

Soft Skills

Programming Languages	Python, R, C/C++, JAVA, ABAP (SAP)
Data Science and Machine Learning	PyTorch, Keras, Tensorflow, Sci-kit Learn
Visualization	Matplotlib, Plotly, Leaflet, QGIS, OpenGL